Ho Chi Minh City University of Technology
Faculty of Computer Science and Engineering

LOGIC DESGIN WITH HDL

Practical session - Semester 242

WEEK 4

Finite State Machine

1 Introduction

1.1 Aims

- Practice in designing digital sequential logic circuit with FSM.
- Understand FSM models and principle of designing logic circuit with FSM.

1.2 Preparation

- Read the laboratory materials before class.
- Review chapter 6 about Finite State Machine.
- Each group prepares at least one laptop with Vivado software installed.

1.3 Report requirements

- Lab exercises will be reviewed directly in class.
- Write report (with circuit/simulation screenshots inserted) in pdf.
- Must have group ID, group member's names and student IDs in the report.
- Compress the report with code files (only .v files) in only one .zip file, name the .zip the group ID (for example: Group1.zip).
- Submit on BK-elearning by deadline.

2 Exercises

2.1 Exercise 1

Design a counter that:

- Counts the following sequence:
  $(1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 8 \rightarrow 10 \rightarrow 11 \rightarrow 12) \rightarrow (1 \rightarrow ...$
- When reset, the counter return value = 1.

- There is a control input:
  - If control = 0, the counter works normally.
  - If control = 1, the counter counts odd number in above sequence only.

Using **Moore** FSM model to implement the counter circuit:

- Draw state diagram that describes the state transition of the counter.
- Write Verilog code to model the designed Moore FSM.

Write a test bench to simulate the counter

Test the counter on FPGA board using Switch (or Button), LEDs, or 7-seg LEDs. Students can generate a low frequency clock to observe the behavior of the circuit on board.

2.2 Exercise 2

Use **Mealy** FSM to design a sequence recognizer that recognizes the sequence '10' or '01' of input.

Ex:

```
Input : 0 → 0 → 0 → 1 → 1 → 0 → 0 → 0 → 1 → 1 → 1 → 1 → 0 → 1
Output: 0 → 0 → 0 → 1 → 0 → 1 → 0 → 0 → 1 → 0 → 0 → 0 → 1 → 1
```

- Output is synchronized with clock (The FSM accepts input at clock edge).
- Output active high when we input 0 then 1 or 1 then 0.
- Draw Mealy state diagram for this circuit.
- Implement the circuit using Mealy model in Verilog.

Write a test bench to simulate the circuit.

Test the circuit on FPGA board using KEY for clock, SW for input and LED for output.

2.3 Exercise 3

Change mode String bit LED circuit.

Use Verilog HDL to model a state machine for a circuit that changes display mode of a bit string.

In initial, LEDs show the default 4-bit random string which is performed by a reset signal, example string: 0011. And buttons in board will set the display mode as follow:

- Button 0: Mode Reset: Show the default 4-bit string on LEDs.

- Button 1: Mode Circular Shift Left Ring : Shift 4-bit string to left in a ring every 1s.

- Button 2: Mode Circular Shift Right Ring: Shift 4-bit string to right in a ring every 1s.
- Button 3: Pause: Pause the current shifting string.

Draw a state diagram to illustrate the designed FSM. Student can use Moore or Mealy model.
Write a test bench to simulate the circuit and test the circuit on the Arty-Z7 board.

Hint: Students should do the following steps:

- Partitioning the design into blocks, may draw a block diagram. Separate the state machine and the string display logic.
- Define the inputs, outputs of the FSM, then design the FSM.
- Modeling the FSM using Verilog HDL. Use the FSM's outputs to control the string display.
- Simulate and test the circuit on board.